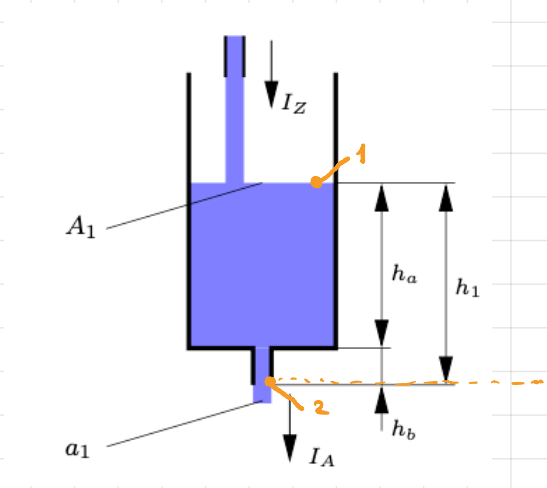

Mass balance:

$$\frac{dm}{dt} = \dot{m}_{in} - \dot{m}_{out}$$

$$\dot{m} = \rho A c \quad , \quad [A] = m^2 , \quad [\rho] = \frac{kg}{m^3} , \quad [c] = \frac{m}{s} , \quad [\dot{m}] = \frac{kg}{s}$$

$$= \rho \cdot \dot{V}$$

$$\Rightarrow \quad \rho \dot{V} = \rho \dot{V}_{in} - \rho \dot{V}_{out} \quad , \quad h_1 \overset{*}{=} h_a + h_b$$

$$V = A_1 \cdot h_a + a_1 \cdot h_b$$

$$\dot{V}_{in} = I_Z \quad , \quad \dot{V}_{out} = I_A$$

$$\Rightarrow \quad \frac{dV}{dt} = I_Z - I_A$$

$$\Leftrightarrow \quad \frac{d}{dt} \left(A_1 h_a^{*} + a_1 h_b \right) = I_Z - I_A$$

$$\Leftrightarrow \quad \frac{d}{dt} \left(A_1 h_1 + (a_1 - A_1) h_b \right) = I_Z - I_A$$

$$\Leftrightarrow \quad A_1 \dot{h}_1 = I_Z - I_A \qquad (h_b = \text{const.})$$

Bernoulli over height h_1: $\quad p_A + \frac{1}{2} \rho v_A^2 + \rho g h_1 = p_a + \frac{1}{2} \rho v_a^2 + \rho g h_2$

$p_A = p_a = p_{atm}, \quad h_2 = 0$

$\Rightarrow \quad \frac{1}{2} v_A^2 + g h_1 = \frac{1}{2} v_a^2$

Assume $A_1 \gg a_1$: $A_1 v_A = a_1 v_a \quad \Rightarrow \quad v_A \approx 0$

$\Rightarrow \quad v_a = \sqrt{2gh_1}$

$\rightarrow \quad I_A = a_1 v_a' = a_1 \mu v_a = a_1 \mu \sqrt{2gh_1}$

$$\Rightarrow \quad \dot{h}_1 = \frac{1}{A_1}(I_z - I_A)$$
$$= \frac{1}{A_1}\left(I_z - a_1 \mu \sqrt{2gh_1}\right)$$

2-Tank:

$$\dot{h}_1 = \frac{1}{A_1}\left(v - a_1 \mu \sqrt{2gh_1} - d\right)$$
$$\dot{h}_2 = \frac{1}{A_2}\left(a_1 \mu \sqrt{2gh_1} - a_2 \mu \sqrt{2gh_2}\right)$$
$$y = h_2$$

System Analysis

Equilibria: $\dot{h}_1 = \dot{h}_2 = 0$

$$\bar{v} = a_1 \mu \sqrt{2g\bar{h}_1} + \cancel{\bar{d}}^{\,0}$$

$$\Rightarrow \quad a_1 \cancel{\mu} \sqrt{2g\bar{h}_1} = a_2 \cancel{\mu} \sqrt{2g\bar{h}_2} \iff \bar{h}_2 = \frac{a_1^2}{a_2^2}\bar{h}_1$$

Linearized model:

$$x := (\Delta h_1, \Delta h_2)^T = (h_1 - \bar{h}_1, h_2 - \bar{h}_2)^T$$

$$y := h_2 - \bar{h}_2$$

$$u := v - \bar{v}$$

$$\frac{d}{dt}(x_1 + \bar{h}_1) = \dot{x}_1 = \frac{1}{A_1}\left(u + \bar{v} - a_1 \mu \sqrt{2g(x_1 + \bar{h}_1)} - d\right)$$
$$\frac{d}{dt}(x_2 + \bar{h}_2) = \dot{x}_2 = \frac{1}{A_2}\left(a_1 \mu \sqrt{2g(x_1 + \bar{h}_1)} - a_2 \mu \sqrt{2g(x_2 + \bar{h}_2)}\right)$$

$$\left.\frac{\partial f_1}{\partial x_1}\right|_{x=0} = \frac{d}{dx_1}\left(-a_1 \mu \sqrt{2g(x_1+\bar{h}_1)}\right)\Big|_{x=0}$$

$$= -a_1 \mu \frac{d}{dx_1}\left(2g(x_1+\bar{h}_1)\right)^{1/2}\Big|_{x=0}$$

$$= -a_1 \mu \frac{g}{\left(2g(x_1+\bar{h}_1)\right)^{1/2}}\Big|_{x=0} = -a_1 \mu \frac{g}{(2g\bar{h}_1)^{1/2}}$$

$$\left.\frac{\partial f_1}{\partial x_2}\right|_{x=0} = 0$$

$$\left.\frac{\partial f_2}{\partial x_1}\right|_{x=0} = \frac{d}{dx_1}\left(a_1 \mu \sqrt{2g(x_1+\bar{h}_1)}\right)\Big|_{x=0} = a_1 \mu \frac{g}{(2g\bar{h}_1)^{1/2}}$$

$$\left.\frac{\partial f_2}{\partial x_2}\right|_{x=0} = -a_2 \mu \frac{g}{(2g\bar{h}_2)^{1/2}}$$

$$\left.\frac{\partial f_1}{\partial u}\right|_{u=0} = 1 \quad , \quad \left.\frac{\partial f_2}{\partial u}\right|_{u=0} = 0$$

$$\frac{\partial f_1}{\partial d} = -\frac{1}{A_1} \quad , \quad \frac{\partial f_2}{\partial d} = 0$$

$$\left.\frac{\partial y}{\partial x_1}\right|_{x=0} = 0 \quad , \quad \left.\frac{\partial y}{\partial x_2}\right|_{x=0} = 1$$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} -\frac{a_1 \mu g}{(2g\bar{h}_1)^{1/2}} & 0 \\ \frac{a_1 \mu g}{(2g\bar{h}_1)^{1/2}} & -\frac{a_2 \mu g}{(2g\bar{h}_2)^{1/2}} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u + \begin{pmatrix} -\frac{1}{A_1} \\ 0 \end{pmatrix} d$$

$$y = \begin{pmatrix} 0 & 1 \end{pmatrix} x$$